

Exercise 5.9

You just bought stock in *FancyTech*. Let random variable μ be the average dollar amount that your *FancyTech* stock goes up or down in a one-day period. At first, you believe that μ is 7.2 dollars with a standard deviation of 2.6 dollars.

- a. Tune and plot an appropriate Normal prior model for μ .
- b. According to your plot, does it seem plausible that the *FancyTech* stock would increase by an average of 7.6 dollars in a day?
- c. Does it seem plausible that the *FancyTech* stock would increase by an average of 4 dollars in a day?
- d. What is the prior probability that, on average, the stock price goes *down*? Hint: `pnorm()`.
- e. What is the prior probability that, on average, your stock price goes up by more than 8 dollars per day?

Exercise 5.10

Continuing with Exercise 5.9, it is reasonable to assume that the daily changes in *FancyTech* stock value are Normally distributed around an unknown mean of μ with a known standard deviation of $\sigma = 2$ dollars.

On a random sample of 4 days, you observe changes in stock value of -0.7 , 1.2 , 4.5 , and -4 dollars.

- Plot the corresponding likelihood function of μ .
- Plot the prior pdf, likelihood function, and the posterior pdf for μ .
- Use `summarize_normal_normal()` to calculate descriptive statistics for the prior and the posterior models.
- Comment on how your understanding about μ evolved from the prior to the posterior based on the observed data.
- What is the posterior probability that, on average, the stock price goes *down*? Hint: `pnorm()`.
- What is the posterior probability that, on average, your stock price goes up by more than 8 dollars per day?

Exercise 5.11

Prof. Abebe and Prof. Morales both recently finished their PhDs and are teaching their first statistics classes at Bayesian University.

Their colleagues told them that the average final exam score across all students, μ , varies Normally from year to year with a mean of 80 points and a standard deviation of 4.

Further, individual students' scores Y vary Normally around μ with a known standard deviation of 3 points.

- a. Prof. Abebe conducts the final exam and observes that his 32 students scored an average of 86 points. Calculate the posterior mean and variance of μ using the data from Prof. Abebe's class.
- b. Prof. Morales conducts the final exam and observes that her 32 students scored an average of 82 points. Calculate the posterior mean and variance of μ using the data from Prof. Morales' class.
- c. Next, use Prof. Abebe and Prof. Morales' combined exams to calculate the posterior mean and variance of μ .